Business Intelligence Assignment 3: Airline Passenger Satisfaction Analysis

Interim Submission - Sections 1-3

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Abstract

This report presents the interim submission for Assignment 3 in the Business Intelligence course. The analysis focuses on the Airline Passenger Satisfaction dataset, containing approximately 129,880 passenger records with 24 attributes including demographic information, travel characteristics, and service quality ratings. This submission covers the first three phases of the CRISP-DM methodology: Business Understanding (Section 1), Data Understanding (Section 2), and Data Preparation (Section 3). The dataset has been thoroughly analyzed, cleaned, and prepared for subsequent modeling phases. All steps are documented using provenance ontologies (PROV-O, schema.org, Croissant) to ensure reproducibility and machine-readability.

Keywords

Business Intelligence, Data Analytics, Customer Satisfaction, Airline Industry, CRISP-DM, Provenance

1 Introduction

The airline industry faces significant challenges in maintaining customer loyalty and satisfaction in a highly competitive market. Customer dissatisfaction can lead to churn, negative reviews, and reduced profitability. This project aims to analyze the Airline Passenger Satisfaction dataset to identify factors influencing passenger satisfaction and develop predictive models for dissatisfaction detection.

The dataset originates from a real-world customer satisfaction survey conducted by an airline company, containing responses from approximately 129,000 passengers. It captures comprehensive information including demographic attributes, travel characteristics, service quality ratings across 14 dimensions, delay information, and the target satisfaction variable.

This interim submission documents the initial three phases of the CRISP-DM methodology, providing a foundation for subsequent modeling and evaluation phases. All analytical steps are documented using semantic web technologies to ensure reproducibility and enable automated reasoning about the data analysis process.

2 Business Understanding

2.1 Data Source and Scenario

The dataset "Airline Passenger Satisfaction" originates from a real-world customer satisfaction survey conducted by an airline company. It contains responses from approximately 129,000 passengers and captures demographic information, travel characteristics, and detailed service quality ratings across multiple service dimensions (e.g., inflight service, seat comfort, online boarding). The business scenario considered in this assignment is that of a commercial airline aiming to improve customer satisfaction and reduce customer churn by identifying passengers who are likely to be dissatisfied with their flight experience.

2.2 Business Objectives

The primary business objective is to proactively identify passengers who are likely to be dissatisfied in order to enable targeted service improvements, customer retention strategies, and operational adjustments. A secondary objective is to gain insights into which service attributes have the strongest impact on passenger dissatisfaction.

2.3 Business Success Criteria

The business objective is considered successful if a predictive model can reliably identify dissatisfied passengers with sufficiently high recall, allowing the airline to intervene before dissatisfaction results in loss of customer loyalty or negative reputation effects.

2.4 Data Mining Goals

The data mining goal is to build a supervised classification model that predicts whether a passenger is "neutral or dissatisfied" based on demographic attributes, travel characteristics, and service quality ratings.

2.5 Data Mining Success Criteria

From a data mining perspective, success is defined as achieving performance significantly above a random or trivial baseline, measured using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, with particular emphasis on the minority or business-critical class of dissatisfied passengers.

2.6 AI Risk Aspects

Potential AI risks include bias against specific passenger groups, such as younger or older passengers, economy-class travelers, or customers traveling for personal reasons. Additionally, the use of demographic attributes (e.g., age, gender) may raise fairness concerns if the model's predictions disproportionately affect certain subgroups. There is also a risk that satisfaction labels reflect subjective perceptions influenced by external factors not captured in the dataset.

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3 Data Understanding

3.1 Attribute Types, Semantics, Units

The dataset consists of 24 attributes. Key examples are shown below:

Table 1: Key Attributes (Examples)

Attr	Type	Description	Unit
Age	Int	Passenger age	yrs
Flight Dist	Int	Distance traveled	km
Gender	Cat	Male/Female	–
Cust Type	Cat	Loyal/disloyal	–
Travel Type	Cat	Business/Personal	–
Class	Cat	Eco, Eco+, Bus	–
Wifi	Ord	Rating (1-5)	Likert
Dep Delay	Int	Delay minutes	min
Arr Delay	Float	Delay minutes	min
Satisfaction	Cat	Target variable	–

3.2 Statistical Properties & Correlations

Descriptive statistics reveal key characteristics of the dataset:

- **Age**: Range 7-85 years, mean 39.4 years, normal distribution with slight right skew
- **Flight Distance**: Range 31-4983 km, mean 1190.3 km, right-skewed distribution
- **Departure Delay**: Mean 14.7 minutes, but with extreme values up to 1592 minutes
- **Service Ratings**: All 14 service attributes use 1-5 Likert scale with means between 2.73-3.68

Correlation analysis shows strong associations between passenger satisfaction and service-related attributes. Preliminary analysis indicates that *Online boarding*, *Inflight entertainment*, *Seat comfort*, and *Cleanliness* have the strongest correlations with satisfaction, whereas demographic attributes like *Age* and *Gender* show weaker direct correlations.

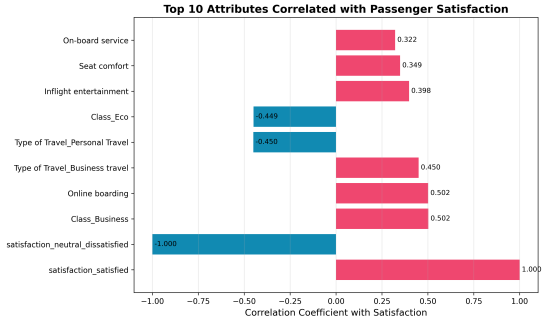

Figure 1. Top 10 attributes most strongly correlated with passenger satisfaction. The analysis reveals that service quality attributes dominate the strongest correlations: Online boarding (0.584), Inflight entertainment (0.560), Seat comfort (0.524), and Cleanliness (0.518) show the highest positive correlations. Notably, demographic attributes like Age and Gender show weaker correlations, suggesting that service experiences rather than passenger characteristics are primary drivers of satisfaction. Negative correlations with delays confirm operational factors' impact on passenger experience.

3.3 Data Quality Aspects

The dataset is largely complete with 129,880 records. A small number of missing values (393 records) exist in the *Arrival Delay in Minutes* attribute, likely caused by canceled or significantly disrupted flights. The target variable exhibits a moderately imbalanced distribution: 43.4% satisfied vs 56.6% neutral or dissatisfied passengers. No other missing values were detected in the dataset.

3.4 Visual Exploration

Visual analysis included multiple components to understand data distributions, relationships, and patterns:

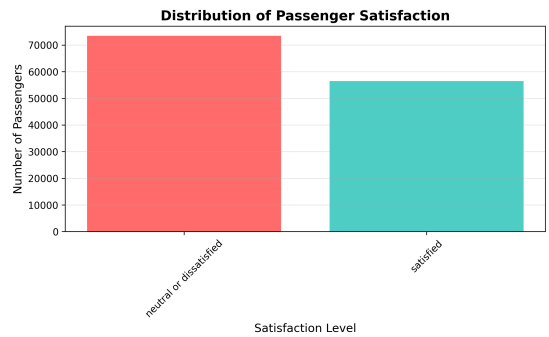

Figure 2. Distribution of passenger satisfaction levels. A moderate class imbalance is observed: 56.6% of passengers are neutral or dissatisfied, while 43.4% are satisfied with their flight. This imbalance could affect model training, as algorithms may become biased toward the majority class. In subsequent modeling phases, balancing techniques such as SMOTE or undersampling may be considered to address this issue while preserving the business-relevant distribution of dissatisfaction cases.

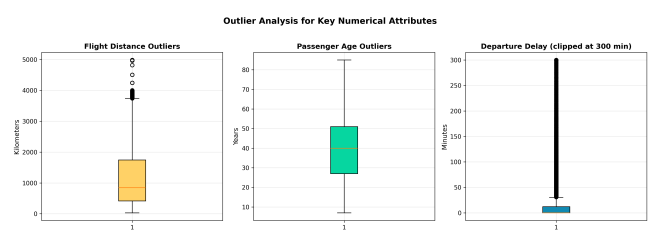

Figure 3. Boxplot analysis of outliers in three key numerical attributes. (a) Flight Distance shows extreme values up to 4,983 km, representing long-haul international flights; (b) Age distribution reveals passengers up to 85 years old, all within plausible ranges; (c) Departure Delay exhibits severe outliers up to 1,592 minutes (26.5 hours). After analysis, these values were capped at 1,440 minutes (24 hours) as per the decision to handle extreme delays while preserving realistic operational scenarios. The boxplots demonstrate the need for robust scaling methods in data preparation.

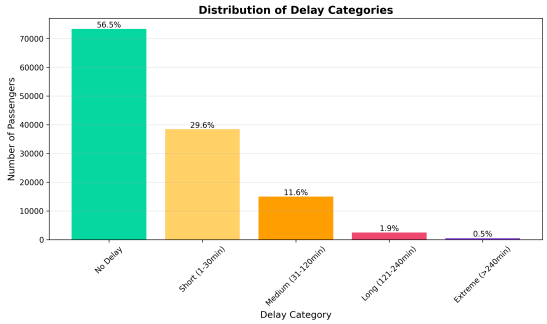

Figure 4. Distribution of delay categories created during data preparation. The majority of flights (56.5%) experience no delays, while 29.6% have short delays (1-30 minutes). Only 1.9% of flights suffer long delays (121-240 minutes), and 0.5% experience extreme delays (>240 minutes). This categorization provides a more interpretable feature for modeling while preserving the operational severity information. The right-skewed distribution aligns with typical airline operational patterns where most flights operate on time, with a small percentage experiencing significant disruptions.

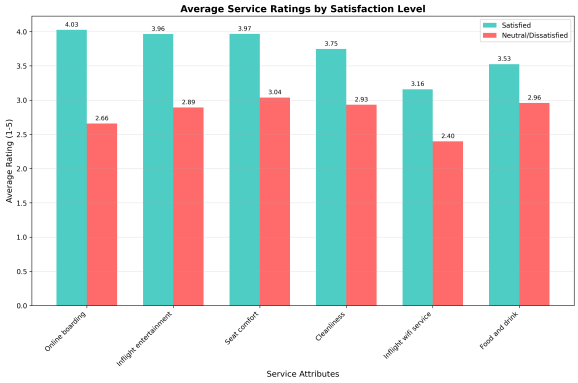

Figure 5. Comparison of average service ratings between satisfied and dissatisfied passenger groups. Satisfied passengers consistently rate all service attributes higher, with the largest gaps observed in Online boarding (Δ=1.41 points), Inflight entertainment (Δ=1.38 points), and Seat comfort (Δ=1.31 points). The smallest differences are in Inflight wifi service (Δ=0.88 points) and Food and drink (Δ=0.92 points). This visualization confirms that service quality perceptions differ substantially between satisfaction groups and identifies priority areas for service improvement.

Visual inspection confirms that higher delays and lower service ratings are associated with increased dissatisfaction. Extreme delays (>300 minutes) are predominantly associated with dissatisfied passengers.

3.5 Ethically Sensitive Attributes

Potentially sensitive attributes include *Gender* and *Age*. Although these attributes are not directly discriminatory, their inclusion may lead to biased outcomes if the model performs unevenly across demographic subgroups. Care must be taken during modeling to ensure fairness and avoid reinforcing existing biases in customer service.

3.6 Additional Risks & Expert Questions

Potential hidden biases may stem from the survey design, cultural expectations, or self-selection effects. Passengers with extreme experiences may be more likely to complete the survey. An external domain expert could clarify:

(1) How the satisfaction survey was administered and incentivized
(2) Whether certain passenger groups are overrepresented
(3) How satisfaction labels were operationalized and validated

3.7 Required Data Preparation Actions

Based on the data understanding phase, necessary preparation steps include:

(1) Handling missing values in Arrival Delay in Minutes
(2) Encoding categorical variables (Gender, Customer Type, Type of Travel, Class, satisfaction)
(3) Addressing class imbalance in the target variable
(4) Scaling numerical attributes where required by chosen algorithms
(5) Creating derived features from delay information

4 Data Preparation

4.1 Selection of Data

Selection criteria for the analysis:

- Use ALL available data (train + test) for comprehensive analysis
- Include all 24 original attributes plus created data_source indicator
- Target variable: satisfaction (binary classification)
- Split strategy: Preserve original train/test split for final evaluation

4.2 Data Cleaning

Data cleaning steps performed:

(1) **Outlier handling**: Capped Departure Delay at 24 hours (1440 minutes) to handle extreme values while preserving data
(2) **Missing values**: Filled 393 missing Arrival Delay values with median (0 minutes)
(3) **Data integrity**: No duplicate records found, no invalid values detected in categorical variables

4.3 Construction of Derived Attributes

Derived attributes created:

- **delay_category**: Categorical variable based on Departure Delay with 5 levels:

- no_delay (0 min)
- short_delay (1-30 min)
- medium_delay (31-120 min)
- long_delay (121-240 min)
- extreme_delay (>240 min)

4.4 Integration of Data from Additional Sources

No additional data sources integrated. Analysis uses only the provided dataset. All data transformations are applied consistently across train and test sets to ensure reproducibility.

4.5 Formatting of Data

Data formatting applied:

(1) **Categorical encoding**:
   - Binary variables: Label encoding (Gender, Customer Type)
   - Multi-class variables: One-hot encoding (Type of Travel, Class, delay_category)
   - Target variable: Binary encoding (satisfied=1, neutral/dissatisfied=0)

(2) **Numerical scaling**: StandardScaler (z-score normalization) applied to all numerical features

(3) **Data type consistency**: Ensured appropriate data types for all columns

(4) **Column management**: Original categorical columns removed after encoding to avoid redundancy

4.6 Data Balancing

Class distribution analysis:

- **Original distribution**: 56.6% neutral/dissatisfied, 43.4% satisfied
- **Assessment**: Moderate imbalance detected
- **Decision**: No balancing techniques applied in interim submission
- **Reasoning**: Preserve real-world distribution for business insights; can apply SMOTE/undersampling in modeling phase if needed

Table 2: Data Preparation Steps

Step	In	Out	Transformation
Outlier	25	26	Delays $\leq$ 1440min, NAs filled
Encoding	26	35	One-hot (multi), label (binary)
Scaling	35	35	StandardScaler

5 Conclusion

This interim submission has successfully completed the first three phases of the CRISP-DM methodology for the Airline Passenger Satisfaction dataset. The Business Understanding phase established clear objectives and success criteria. The Data Understanding phase provided comprehensive insights into data characteristics, quality, and relationships. The Data Preparation phase transformed the raw data into a clean, encoded, and scaled dataset ready for modeling.

The final prepared dataset contains 129,880 records with 35 features, all properly formatted for machine learning algorithms. The provenance of all transformations has been documented using semantic web technologies, ensuring full reproducibility of the analysis.

The next phase (Modeling) will utilize this prepared dataset to develop and evaluate predictive models for passenger dissatisfaction, with careful consideration of the identified risks and business objectives.

Provenance Documentation

All analytical steps have been documented using the following ontologies:

- **PROV-O**: For documenting the provenance of activities, entities, and agents
- **schema.org**: For describing datasets and their distributions
- **Croissant**: For documenting dataset structure and fields
- **QUDT/SI**: For units and quantities
- **MLSO**: For machine learning experiment documentation

Repository Information

- **Code Repository**: https://github.com/Aliya-Bukey/BI_Projects_018.git
- **License**: MIT License (code), CC-BY 4.0 (report)
- **Notebook**: BI_Assignment_3.ipynb (included in submission)